

Course Syllabus

STI VISION: To be the leader in innovative and relevant education that nurtures individuals to become competent and responsible members of society.		STI INSTITUTIONAL OUTCOMES:	
STI MISSION: We are an institution committed to provide knowledge through the development and delivery of superior learning systems.		Character (IO1): An STler is a person of character. An STler takes responsibility for his/her actions, treats people with respect, and lives with integrity.	
We strive to provide optimum value to all our stakeholders - our students, our faculty members, our employees, our partners, our shareholders, and our community.		Critical thinker (IO2): An STler is a critical thinker. An STler challenges and analyses all information through sound questioning and is unafraid to push for creative ideas.	
We will pursue this mission with utmost integrity, dedication, transparency, and creativity.		Communicator (IO3): An STler communicates to understand and be understood. An STler discerns the value of information read or heard and effectively expresses his/her own emotions when sharing information, may it be spoken or written.	
		Change-adept (IO4): An STler is change-adept. An STler can adjust, adapt, and reinvent continuously to changing circumstances. An STler believes in letting go of the old and embracing the new to achieve his/her fullest potential.	
SERIAL NUMBER: IT2202	COURSE TITLE: COMPUTER GRAPHICS PROGRAMMING		CREDIT: 2 lec 1 lab (2 hours lecture and 3 hours laboratory per week)
COURSE DESCRIPTION:	This course covers the introduction to the mathematics of computer graphics as well as how they are applied to created 3D graphics programs.		
PREREQUISITE:	Computer Programming 2		
COURSE OUTCOMES:	After successful completion of this course, the student should be able to: (PI1-ITPO01, PI2-ITPO02, PI3-ITPO05, PI4-ITPO07, PI5-ITPO09, PI6-ITPO10) CO1. Examine concepts and data structures to represent and manipulate geometry; CO2. Develop 2D and 3D computer graphics programs; and CO3. Implement various approaches and techniques in creating interactive applications.		
MANDATED BOOK:	None		
REFERENCES:	1. de Fine, M. (2018). <i>Python 2.6 graphics cookbook</i> . Packt Publishing. 2. Kelly, S. (2019). <i>Python, Pygame, and Raspberry Pi game development</i> . Apress. 3. Korites, B. (2018). <i>Python graphics: A reference for creating 2D and 3D images</i> . Apress. 4. Orland, P. (2019). <i>Math for programmers: 3D graphics, machine learning, and simulations with Python</i> . Manning. 5. Stemkoski, L. & Pascale, M. (2021). <i>Developing graphics frameworks with Python and OpenGL</i> . CRC Press.		

COURSE REQUIREMENTS:	Class Participation (<i>Recitation, Seatwork, Quizzes</i>) Major Examinations Performance Tasks (<i>Laboratory Exercises, Projects</i>)																		
GRADING SYSTEM:	The following percentage distribution shall be followed: <table><tr><td>Prelims</td><td>20%</td></tr><tr><td>Midterms</td><td>20%</td></tr><tr><td>Pre-finals</td><td>20%</td></tr><tr><td><u>Finals</u></td><td><u>40%</u></td></tr><tr><td></td><td>100%</td></tr></table> The following are the recommended periodical grade components for this course: <table><tr><td>Quizzes</td><td>20%</td></tr><tr><td>Performance Tasks/Laboratory Exercises</td><td>30%</td></tr><tr><td><u>Major Examination</u></td><td><u>50%</u></td></tr><tr><td></td><td>100%</td></tr></table>	Prelims	20%	Midterms	20%	Pre-finals	20%	<u>Finals</u>	<u>40%</u>		100%	Quizzes	20%	Performance Tasks/Laboratory Exercises	30%	<u>Major Examination</u>	<u>50%</u>		100%
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We'd be glad to hear from you. For questions or feedback on this course, feel free to email us through student.feedback@sti.edu																			

Course Outline

Learning Objectives (LO)		Week	LEC Hours	LAB Hours	TOPICS	Teaching and Learning Activities	Assessment Tasks
1	Describe the core concepts used in computer graphics (CO1)	1-2	4	6	Introduction to Computer Graphics Core Concepts The Graphics Pipeline	Lecture-Demonstration	01 Seatwork 1
2	Compare the stages in the graphics pipeline (CO1)						01 Laboratory Exercise 1
3	Set up a development environment (CO2)						
4	Draw graphics in windows (CO2)	3-4	4	6	Pygame and OpenGL Windows, Points, and Shapes Uniform Data Interactivity	Lecture-Demonstration	02 eLMS Quiz 1
5	Create animations and interactive applications (CO2)						02 Laboratory Exercise 1 02 Performance Task 1
		5			PRELIMINARY EXAMINATION		Pen and Paper Test
6	Perform vector addition and scalar multiplication (CO1)	6-7	4	6	Vectors and Matrices Vectors Matrices	Lecture-Demonstration	03 Seatwork 1
7	Add, subtract, and multiply matrices (CO1)						03 Laboratory Exercise 1
8	Create matrices representing geometric transformations (CO1)	8-9	4	6	Geometric Transformations Scaling, Rotation, and Translation Projections	Lecture-Demonstration	04 eLMS Quiz 1
9	Incorporate matrix objects into the rendering of interactive 3D scenes (CO2)						04 Laboratory Exercise 1 04 Performance Task 1
		10			MIDTERM EXAMINATION		Pen and Paper Test
10	Create the structure of a 3D graphics framework (CO2)	11-13	6	9	A Scene Graph Framework The Class Structure 3D Objects Geometry and Material Objects Rendering	Lecture-Demonstration Group Activity	05 Laboratory Exercise 1
11	Develop classes to generate geometric data (CO2)						05 eLMS Quiz 1
12	Render objects in the framework (CO2)						05 Performance Task 1
		14			PRE-FINAL EXAMINATION		Pen and Paper Test

13	Apply textures to the surfaces of geometric shapes (CO3)	15-17	6	9	Textures, Lights, and Shadows Textures Lights Shadows	Lecture-Demonstration	06 Laboratory Exercise 1
14	Create light objects of different types (CO3)					Group Activity	06 eLMS Quiz 1
15	Add shadow rendering capabilities to the framework (CO3)						06 Performance Task 1
		18			FINAL EXAMINATION		Pen and Paper Test

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